Football Object Detection Sample Dataset

1. Project Overview

This project demonstrates the object detection annotation capabilities of Annomax AI using football match images. The dataset showcases accurate bounding box placement, consistent object classification and quality assurance practices for computer vision applications.

Project Name: Football Object Detection Sample Dataset
Organization: Annomax AI
Task Type: Image Annotation
Annotation Method: Bounding Box
Dataset Size: 20 images (planned)
Dataset Status: Portfolio Demonstration

2. Project Objectives

- Demonstrate accurate bounding box annotation of objects in football match images.
- Identify and classify players and relevant football objects.
- Maintain consistent annotation guidelines across all images.
- Apply quality checks to identify missed objects, incorrect labels and inaccurate bounding boxes.
- Prepare structured annotations for object detection and sports analytics workflows.

3. Object Classes

The dataset uses the following object classes:

Class ID	Class Name	Description
0	player	Outfield football players
1	goalkeeper	Players acting as goalkeepers
2	referee	Match officials visible on the pitch

Class ID	Class Name	Description
3	ball	The football used during the match
4	goalpost	Visible goal frames and posts

Class IDs are used for documentation and consistent labeling. The final exported IDs must follow the class configuration of the annotation platform.

4. Annotation Guidelines

- Annotate each visible target object with its own bounding box, even when multiple players are close together or overlap.
- Overlapping bounding boxes are allowed when objects are close to each other or partially occlude one another.
- Draw each bounding box as tightly as possible around the visible boundaries of its target object without including unnecessary parts of nearby objects.
- Do not merge multiple players into a single bounding box.
- When one player partially blocks another, annotate the visible portion of each player according to the project guidelines.
- If an object is heavily occluded and its boundaries are difficult to determine, annotate the clearly identifiable visible portion and flag uncertain cases for review when necessary.

5. Dataset Structure

The recommended folder structure is:

```
football-object-detection-dataset/
├── README.md
├── classes.txt
├── images/
│   ├── image_001.jpg
│   ├── image_002.jpg
│   └── ...
├── annotations/
│   └── annotations.json
├── preview/
│   └── annotated_samples/
└── qa_report.xlsx
```

The actual folder structure may vary depending on the annotation platform and export format.

6. Annotation Format

The dataset may be exported in COCO JSON or YOLO format, depending on project requirements.

- **COCO JSON:** Stores image information, category definitions and bounding box coordinates in a structured JSON file.
- **YOLO:** Stores class IDs and normalized bounding box coordinates in text files, usually one label file per image.

The final export should be validated to ensure that class IDs, image references and bounding box coordinates are correct.

7. Quality Assurance

The dataset should be reviewed for:

- Correct object class assignment
- Accurate bounding box placement
- Missing visible objects
- Duplicate annotations
- Incorrect or inconsistent labels
- Excessive background inside bounding boxes
- Invalid or incomplete annotation records
- Consistent image and label file naming
- Correct export structure and coordinate values

A second review should be performed before finalizing the dataset whenever possible.

8. Potential Applications

This type of annotated data can support:

- Football player detection
- Sports object tracking
- Player counting and positioning
- Football match analysis
- Sports video analytics
- Computer vision model development
- Automated sports content analysis

This sample demonstrates annotation methods and does not claim that a trained or validated AI model has been developed.

9. Data Source and Usage

Images should be collected from original sources that permit the intended use. Source information and applicable license details should be documented where available.

This dataset is prepared for portfolio demonstration and service evaluation. It is not represented as client work or production-ready training data.

Redistribution of source images and annotations must comply with applicable image licenses and permissions.

10. Organization

Annomax AI

AI data annotation, data processing and quality assurance services for businesses and AI development teams.

Website: www.annomaxai.com

Email: sales@annomaxai.com

11. Disclaimer

This is a portfolio demonstration project created to showcase image annotation workflows, labeling consistency and quality assurance practices. Actual dataset performance and suitability depend on the specific requirements of each project.